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“Remote management over Ethernet is still to be widely deployed in the industrial space”

Use of control systems, machines and information technology to optimise a system leads to two of the most important goals for a production facility—productivity and efficiency. Obviously, the electronics behind the automation systems plays a very important role in achieving those goals.

Brian Brown, vice president of marketing, embedded computing, Emerson Network Power, spoke to Dilin Anand of EFY Group about the latest in product development for the industrial automation space

Computer-on-module (CoM) is something that everybody involved in automation is talking about. Could you explain how CoMs help reduce the time to market and lower risk as well?

COMs are still growing in popularity and are widely adopted across medical, gaming and industrial automation. Their benefits are many: They are widely available. They reduce the risk and time by avoiding the need to design high-speed CPU and memory circuitry; enable product flexibility using a single carrier design to support multiple CoMs; and provide a level of future proofing. In addition, they support x86, Power and ARM architectures. Using a CoM approach can help engineers to solve industrial automation challenges quickly. They can concentrate on their application-specific carrier and software, which is a much easier task than a complete solution including a CPU design.

What are the main challenges faced by engineers while working with CoMs?

This sort of hybrid program needs careful coordination. The COM often has to initialise devices on a custom carrier and therefore requires specific BIOS or board support package from the COM vendor.

In many cases, using the combination of a COM and a carrier is practical compared to a complete custom de-

sign. That's because in hybrid CoM approach, where the design is split between an off-the-shelf processor and a customer-specific carrier, it is possible (and in some cases cost-effective) for a company like us to provide a complete custom solution. Each application is slightly different and you need experts in diverse disciplines such as life cycle management, thermal simulation, shock and vibration resistance, packaging and throughput.

How can engineers enhance the energy efficiency of their designs without much efforts?

Energy efficiency requires more of an evolutionary than revolutionary approach. With each new generation of our processor boards, we are typically able to extract more performance for the same power.

In some industrial applications it makes sense to plan a refresh cycle to use the more energy-efficient boards that are largely backwards compatible from an interface and form factor perspective. At the same time, new boards can enable incremental power savings features.

In a portable environment such as a handheld industrial test gadget, it's highly desirable for the device to last a complete shift on a single charge, and so enabling low-power 'drowse' modes when not in use is sensible. If it takes the device 0.5 second to come back up to operational speed, then that is likely acceptable.